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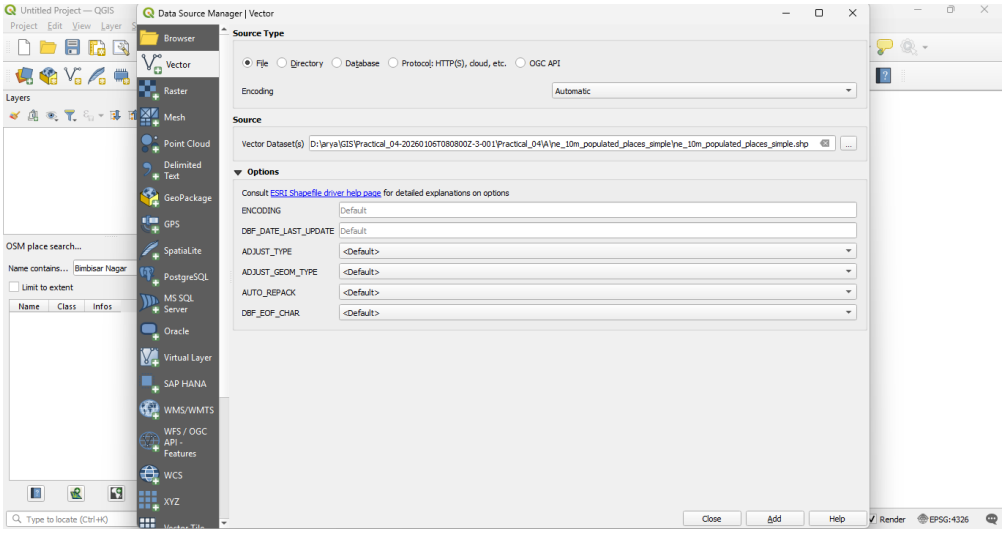
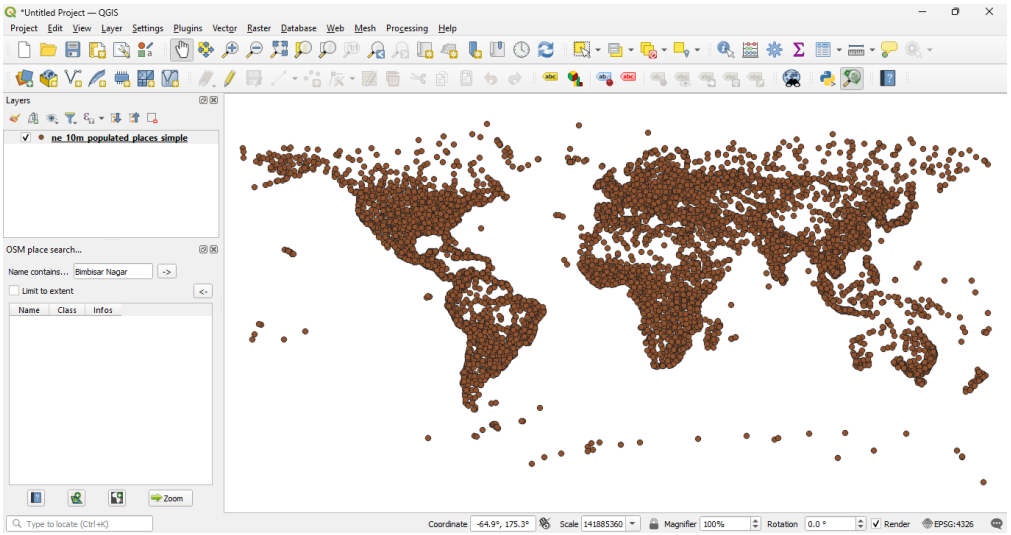
Q] Working with Terrain Data attributes and hill shapes.

Steps:

- Start a new project.
- Go to Layer → Add Layer → Add Vector Layer
- Select "\GIS_Workshop\Practicals\Practical_04\A\Data\ne_10m_populated_places_simple.zip"





- Right click on Layer in Layer Panel → ***Open Attribute Table***.

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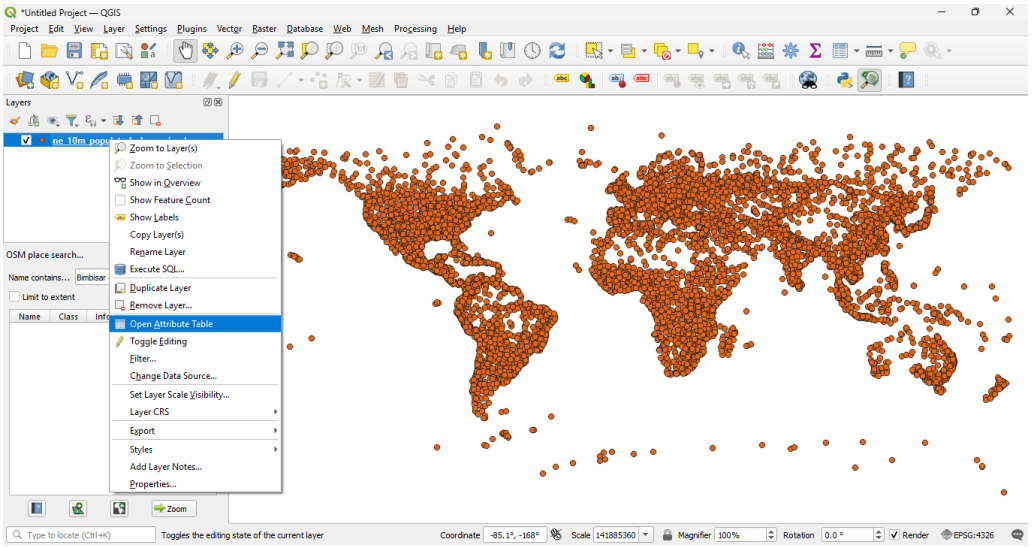
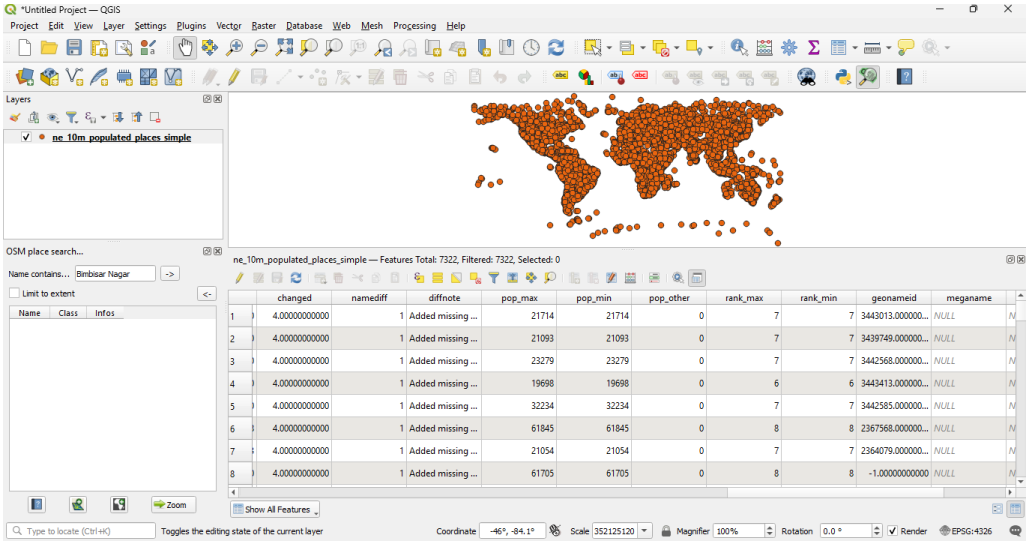
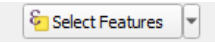


- Explore various attributes and their values in the Attribute table.
- To find the Place with ***maximum population click on "pop_max" file***



	changed	namediff	diffnote	pop_max	pop_min	pop_other	rank_max	rank_min	geonameid	meganame
1	4.00000000000	1	Added missing ...	21714	21714	0	7	7	3443013.000000...	NULL
2	4.00000000000	1	Added missing ...	21093	21093	0	7	7	3439749.000000...	NULL
3	4.00000000000	1	Added missing ...	23279	23279	0	7	7	3442568.000000...	NULL
4	4.00000000000	1	Added missing ...	19698	19698	0	6	6	3443413.000000...	NULL
5	4.00000000000	1	Added missing ...	32234	32234	0	7	7	3442585.000000...	NULL
6	4.00000000000	1	Added missing ...	61845	61845	0	8	8	2367568.000000...	NULL
7	4.00000000000	1	Added missing ...	21054	21054	0	7	7	2364079.000000...	NULL
8	4.00000000000	1	Added missing ...	61705	61705	0	8	8	-1.00000000000	NULL

- On clicking the Select feature using expression button the following window will appear.
- Enter ***pop_max>100*** and pop_max button 

to get all the places with population between 100 and 10000.

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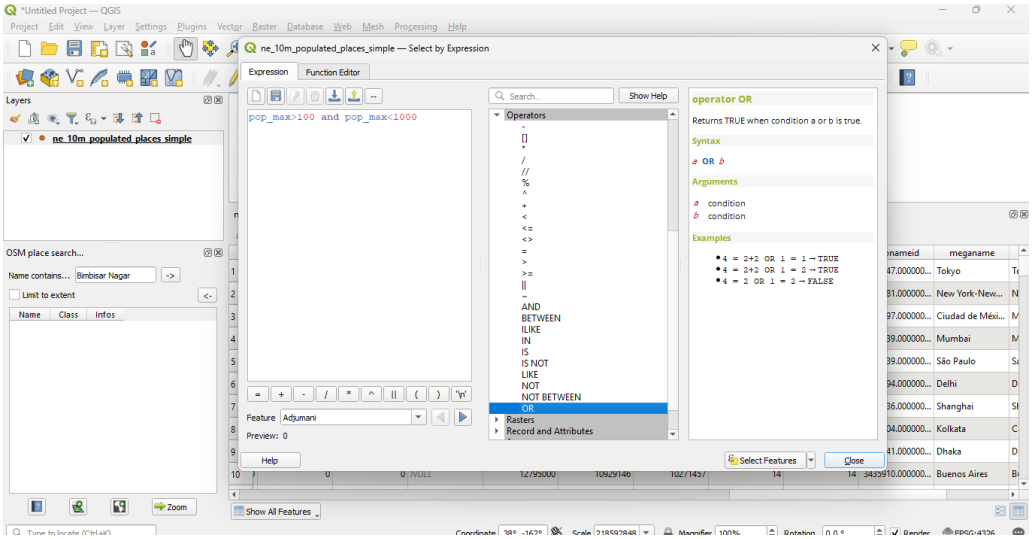
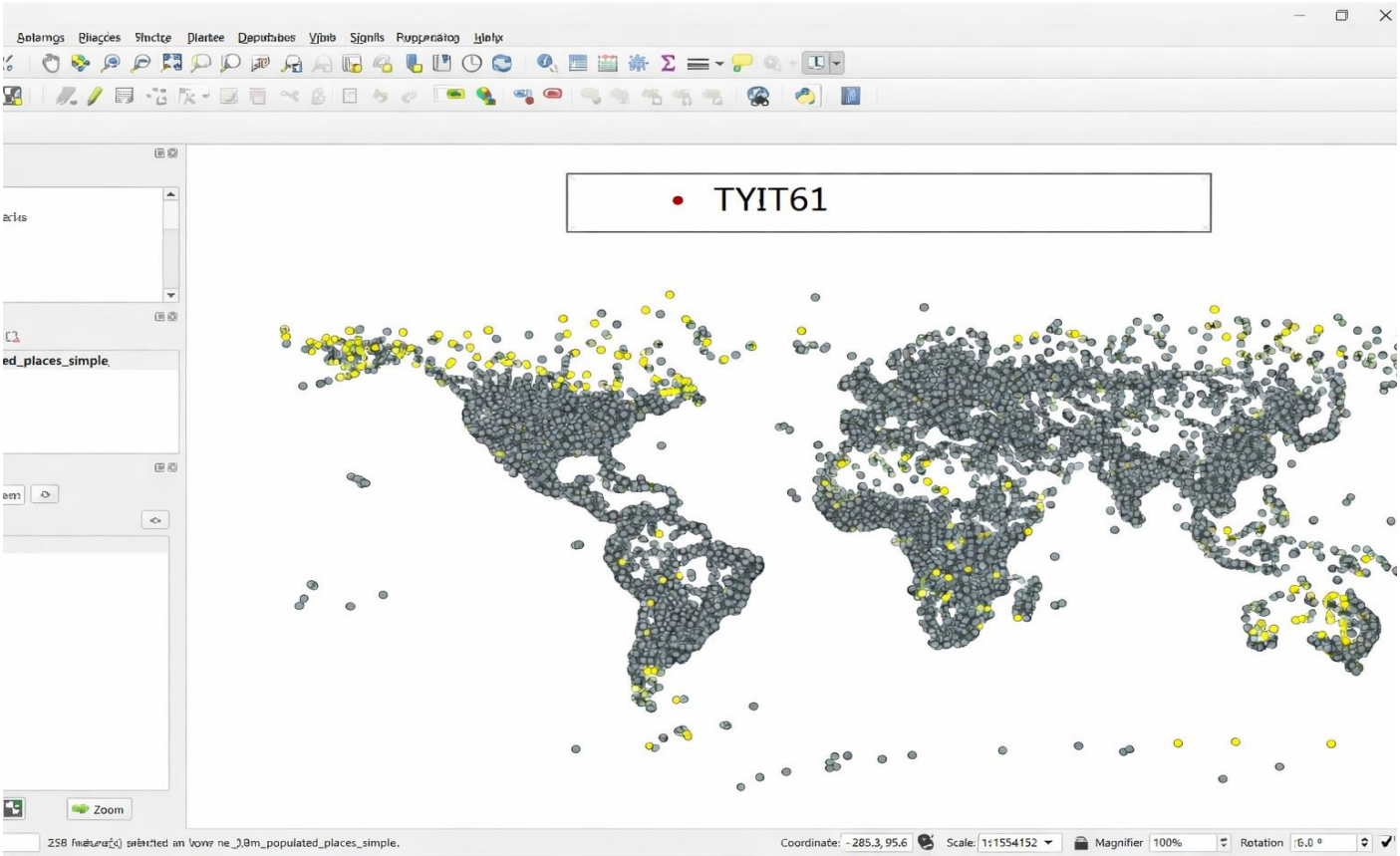


- The places matching the criteria will appear in different color.



- Different queries can be performed using the dataset.
- Try this

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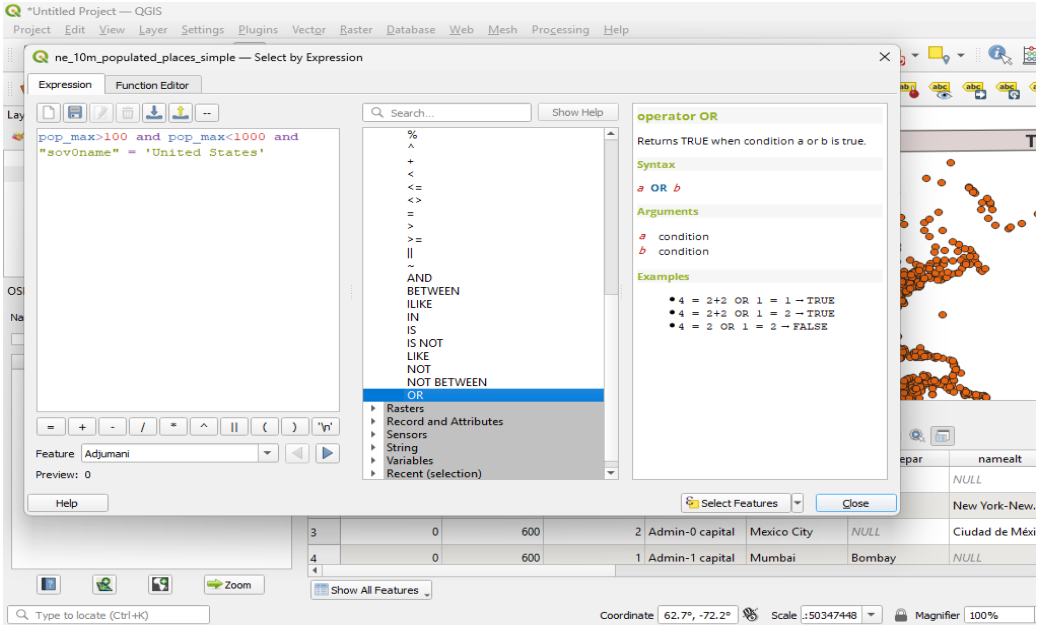

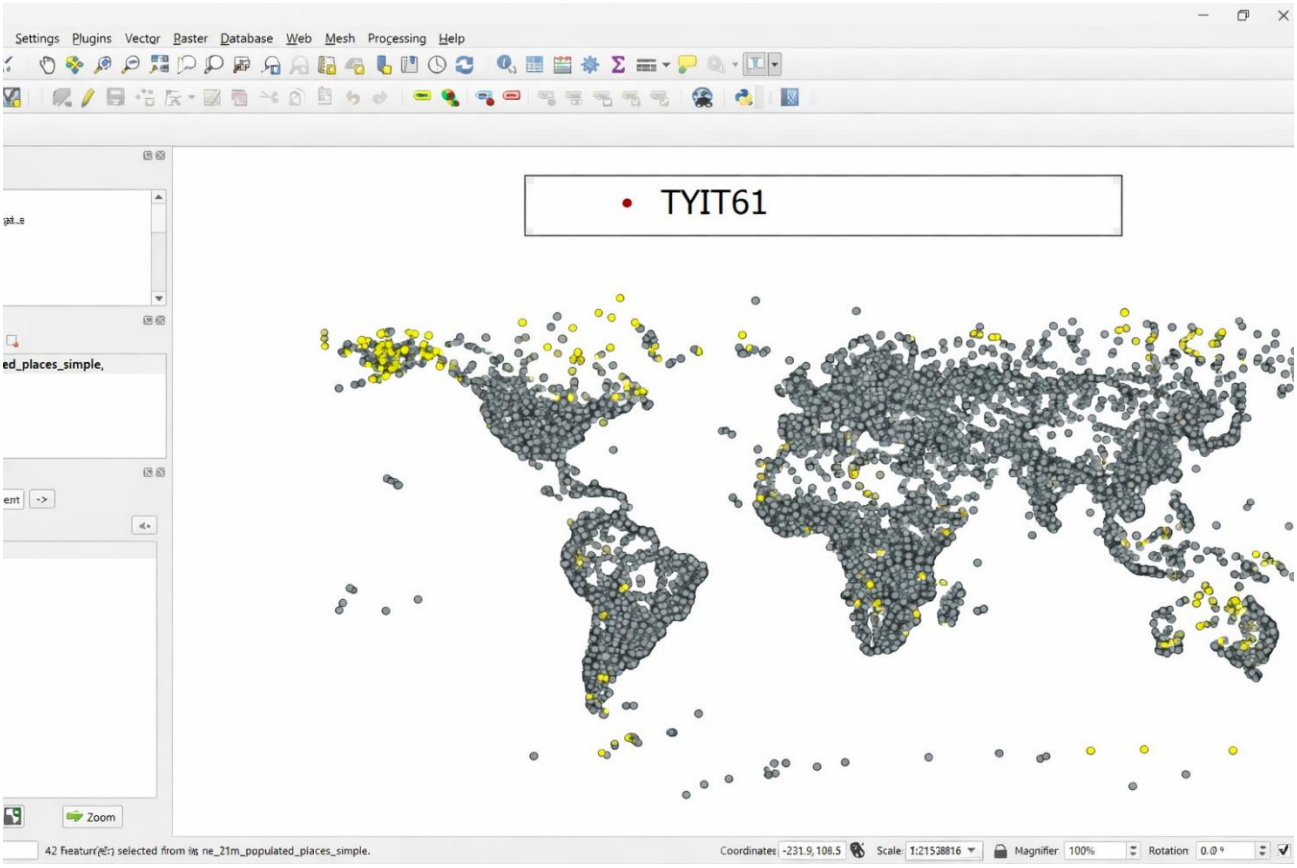

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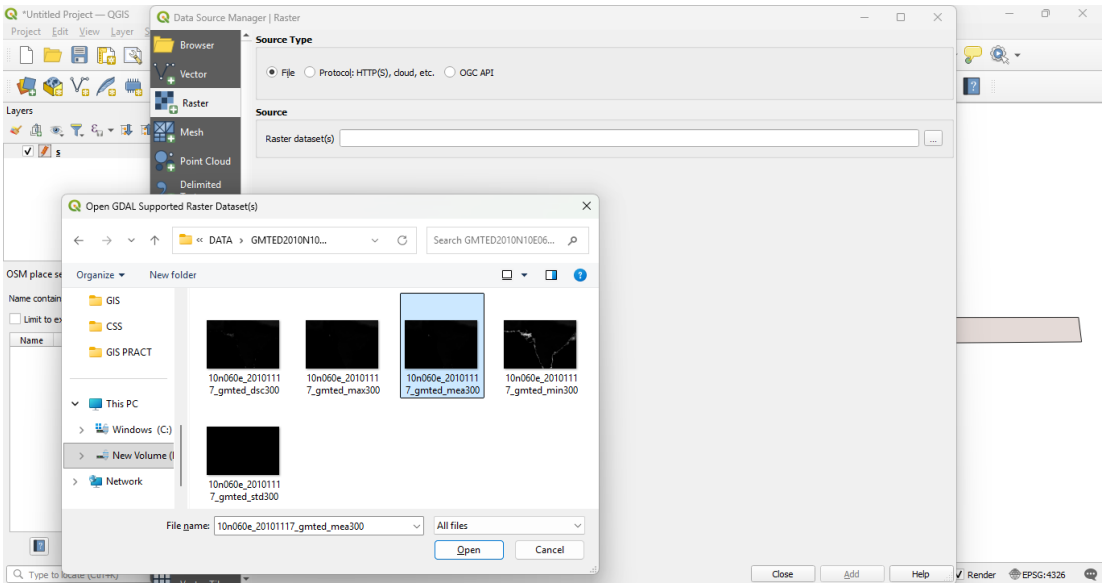
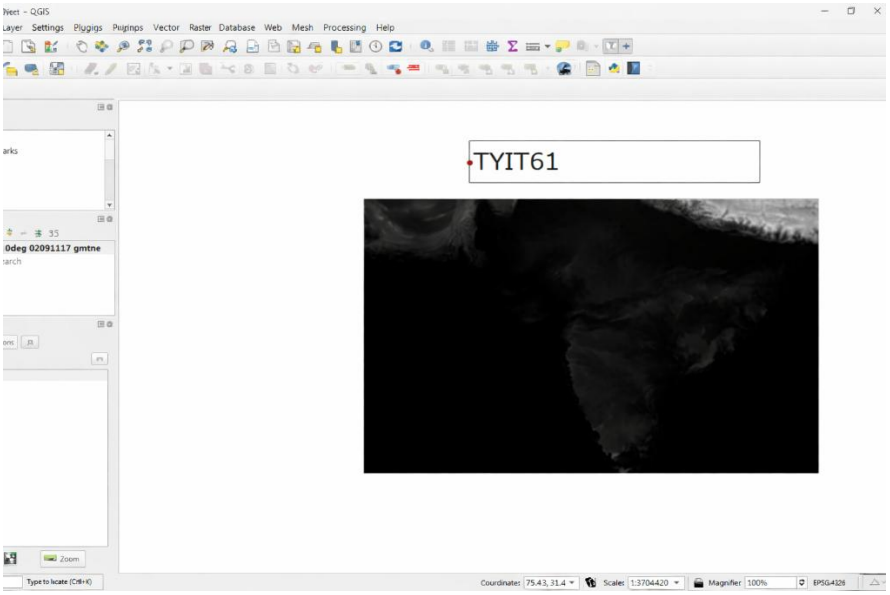
Terrain Data and Hill shade analysis:

A terrain dataset is a multiresolution, TIN-based surface built from measurements stored as features in a geodatabase. Terrain or elevation data is useful for many GIS Analysis like, to generate various products from elevation data such as contours, hill shade etc.

Steps:

- Go to Layer → Add Raster Layer → select "10n060e_20101117_gmted_mea300.tif", from Data folder





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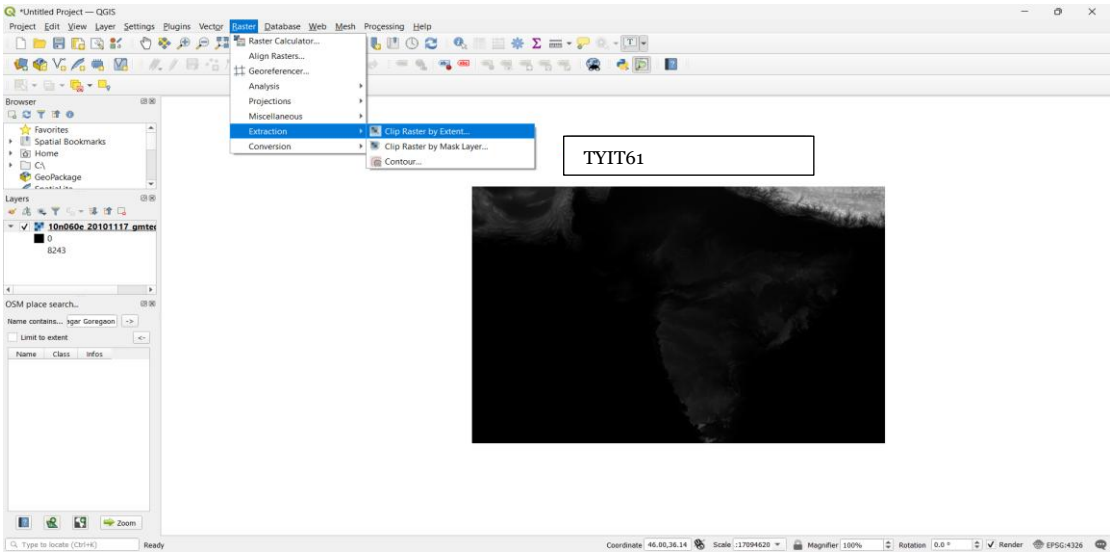
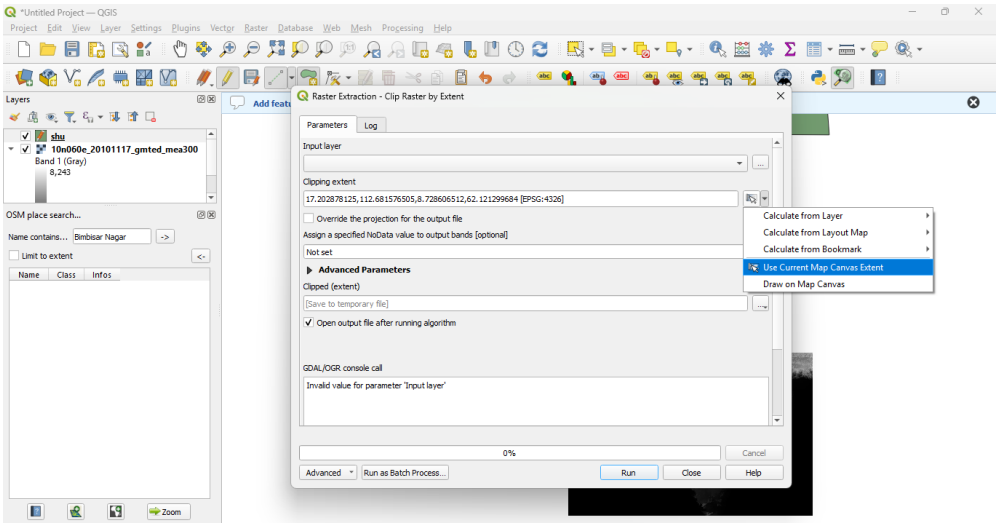
- Go to Raster → Extraction→ Clip Raster by Extent



- Select the raster layer (if project contains multiple layers).
- Select the clipping area by selecting the option Use Canvas Extends if the visible part of map is to be selected or manually select an area on canvas by using Select Extent on Canvas.



- Select the location and file name for storing clipped raster layer

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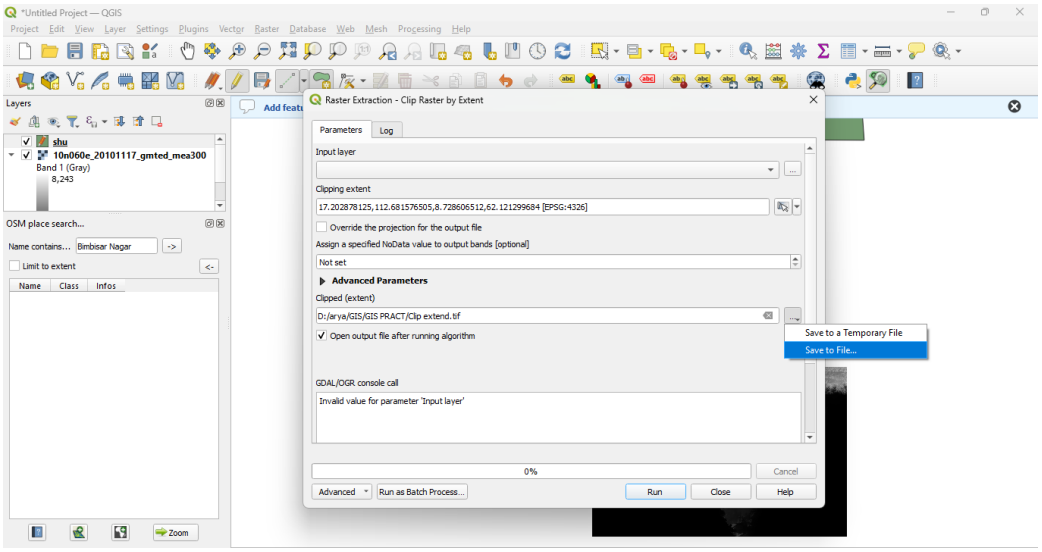
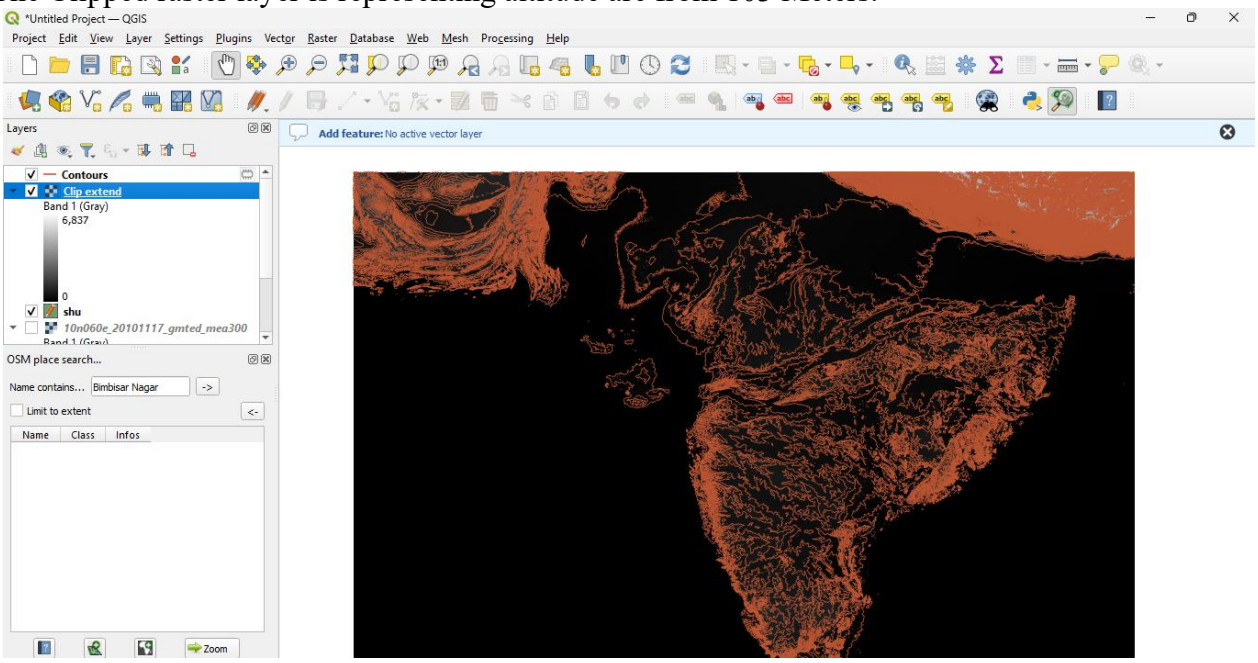


- Press RUN.
- Deselect the original layer and keep the clipped one.
- The Clipped raster layer is representing altitude are from 103 Meters.



- Counter lines are the lines on a map joining points of equal height above or below sea level. A contour interval in surveying is the vertical distance or the difference in the elevation between the two contour lines in a topographical map
- To derive counter lines from given raster.
- Go to Raster → Extraction→ Contour

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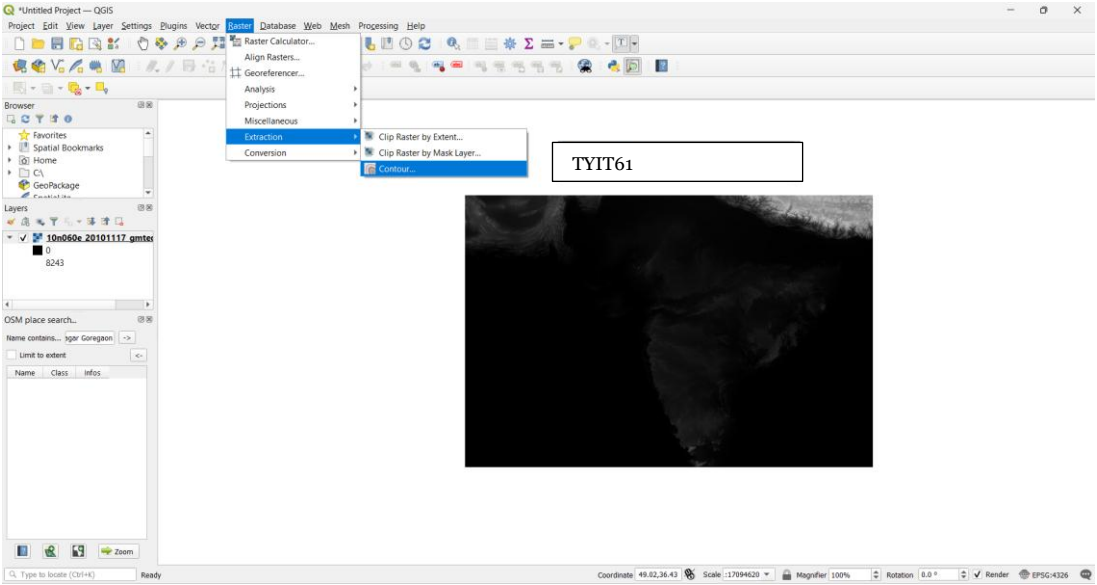
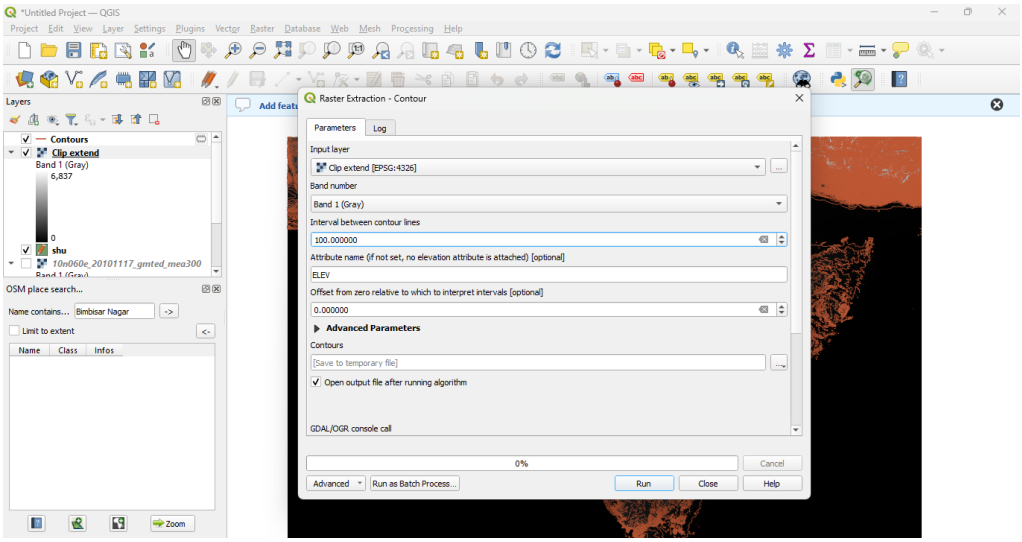


- The Contour configuration window will appear



- Select the input raster layer name. Set contour interval 100.00 meters, select the output file name & location and check the option to add output file to project after processing.
- Press "RUN".

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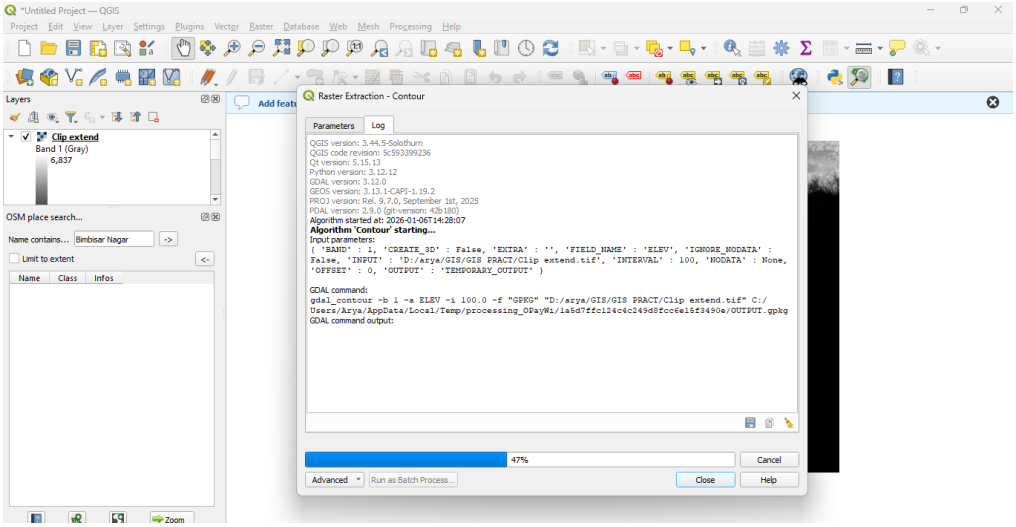
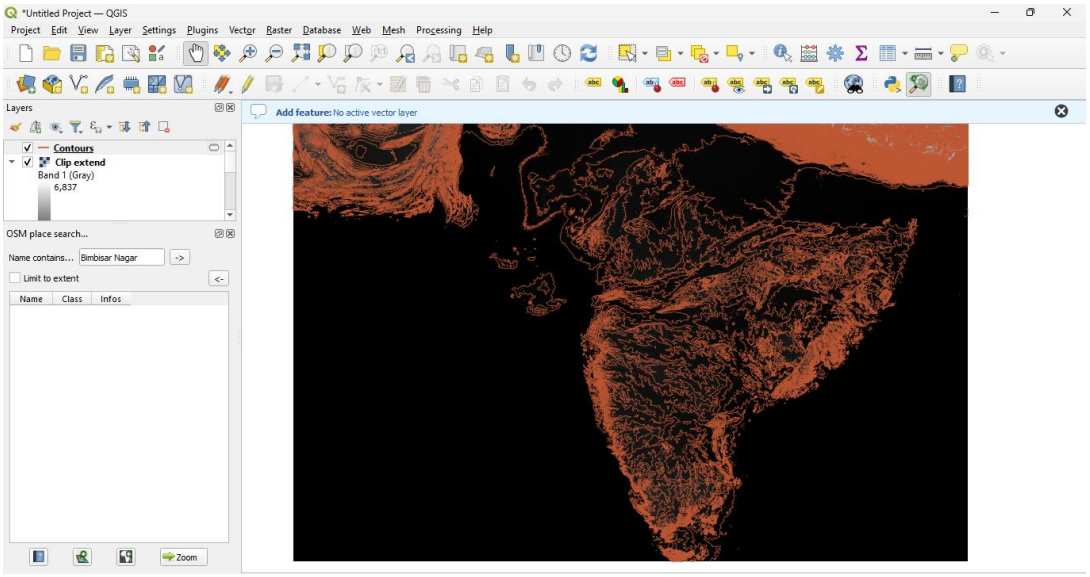


- The contour layer will appear like this



- Go to properties of Contours -> Labels -> Single Labels -> Change the color blue -> Change the size to required tables.

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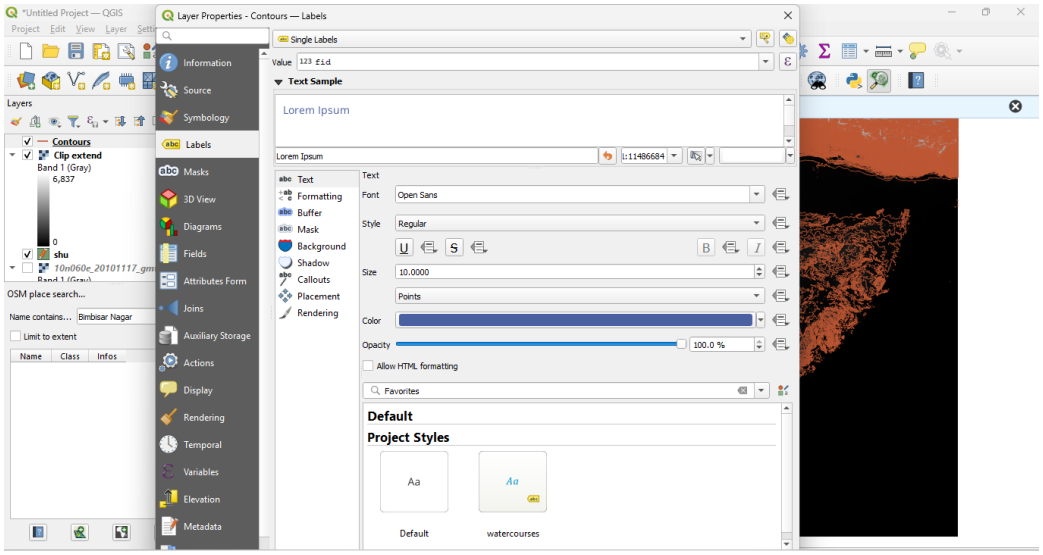
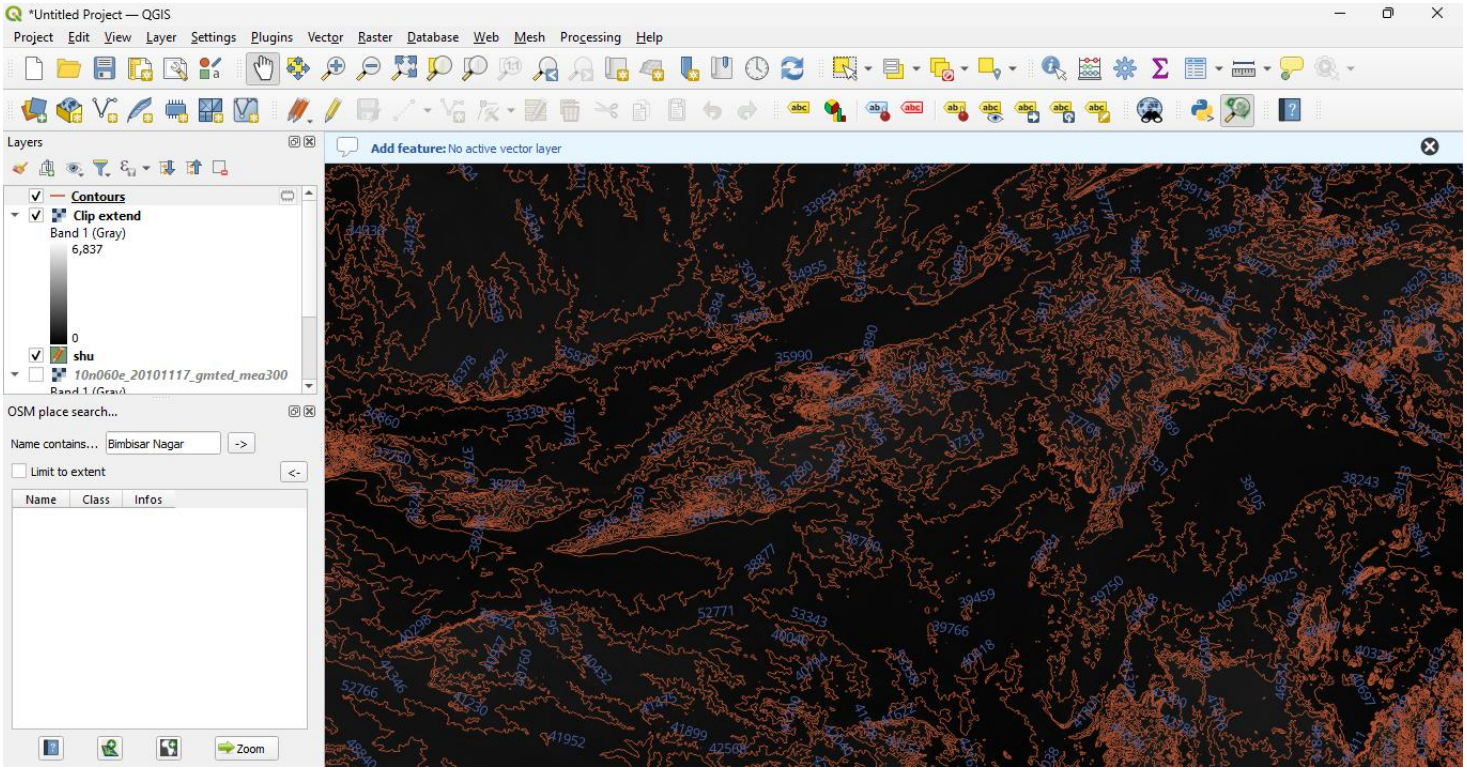


- The Final Map will look like this



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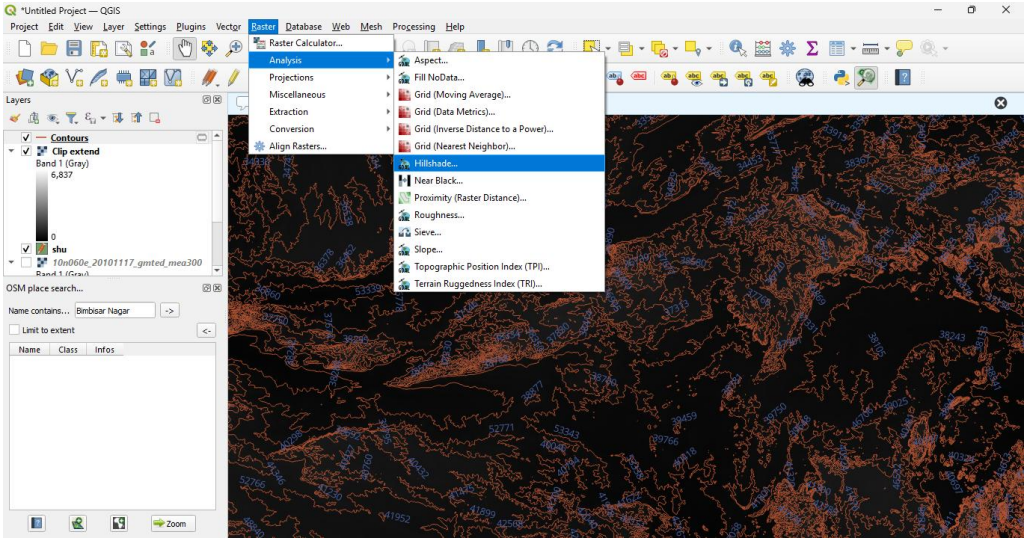
What is a Hillshade?
A Hillshade is a grayscale 3D representation of the surface, showing the topographical shape of hills and mountains using shading (levels of gray) on a map, just to indicate relative slopes, mountain ridges, not absolute height.

Steps:

- For Hill Shade surface analysis.
- Go to Plugin → Install Georeferenced GADL.
- After successful installation of plugin Go to Raster → Analysis → Hill Shade



- Select the input raster layer, select file name and location for storing Hill Shade output file. Select the input layer as the clip extend that we created.

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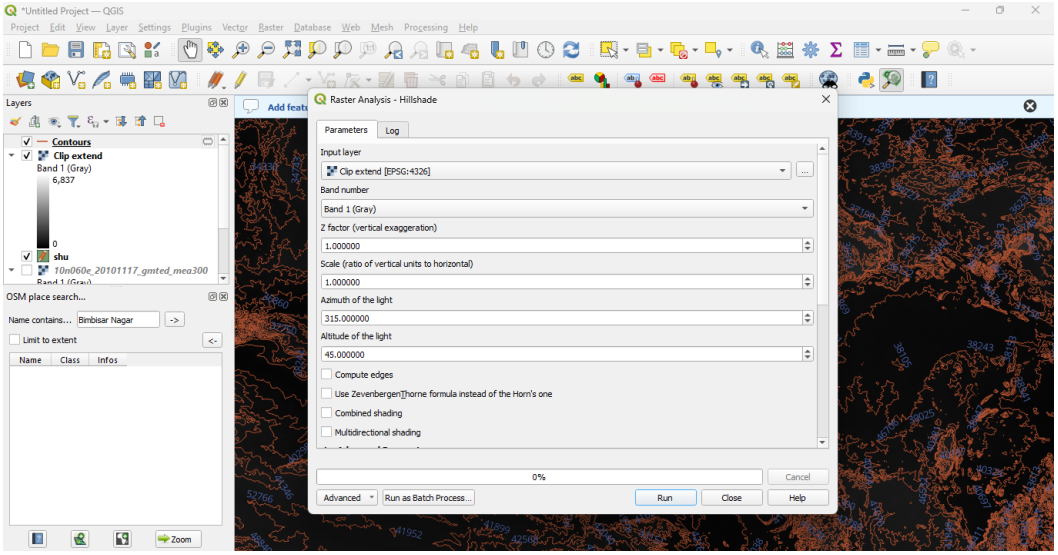
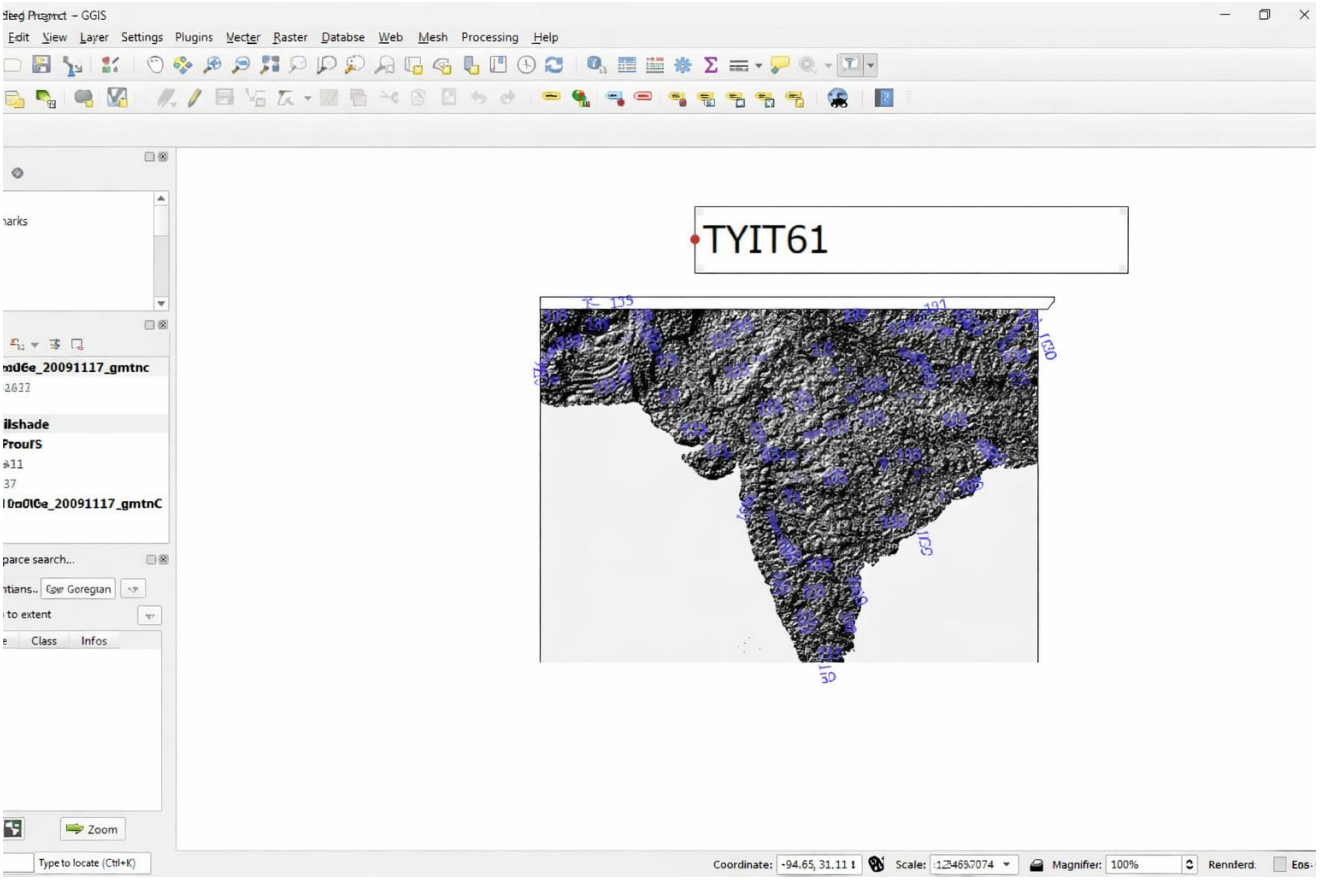


- Press "RUN" and Close the Hill Shape Dialog window.
- After Raster styling the Output will appear like this.



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